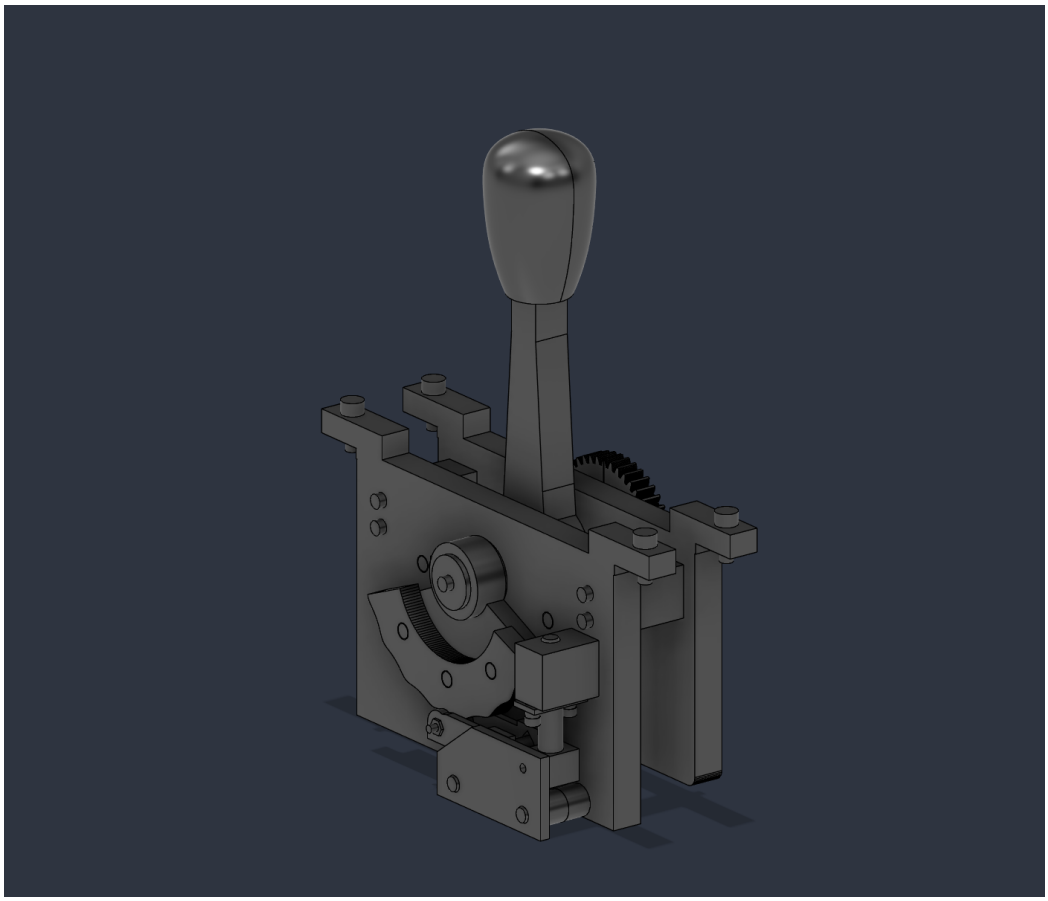
01 / Opposite-side lever

Assembly and cam-service guide

Cam and follower on the left; geared potentiometer on the right. The mechanism hangs from the top rails. No top panel is included. Preserve the pre-gear v6 build for comparison.



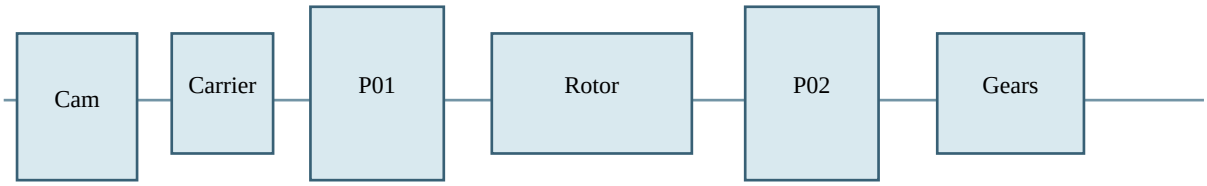
Actual Fusion model: cam-service side. The large side plates carry the main pivot and rail loads; the outer cassette carries the follower.

Travel	Sensor drive	Cam options
120 degrees mechanical (+/-60)	48:24 gears, pot rotates twice lever angle	3, 5, 7 or 8 gates over 116 degrees

Prototype: pot travel, spring solid height, print fit, strength and wear still require physical checks. CAD interference checks do not establish a load rating. The review records and validation notes accompany the geometry.

02 / Parts and axial layout

LEFT: service access RIGHT: sensor drive



Schematic axial order; not a dimensional drawing

Hardware for one geared build	Qty	Use
M3 x 20 cap screw	16	Braces 8, springblock 2, cassette 2, grip 2, roller 1, preload 1
M3 x 16 cap screw	3	Stops 2, rotor clamp 1
M3 x 12 cap screw	7	Gear clamps 2, pot panel 2, cup guides 2, temporary service 1
M3 x 10 / M3 x 8	4 / 3	Cam 3 + carrierclamp 1 / shaftjoin 2 + cap 1
M3 plain nut / CNC M3 inserts	1 / 32	Roller axle / all specified insert pockets
M5 x 14 / 2020 M5 T-nuts	4 / 4	Top rail attachment
623ZZ bearings	2	Follower roller, 3 x 10 x 4 mm each
Spring / measured pot	1 / 1	Wire 1, OD9, free 20 / body 11.2 x 9.5 x 6.6 mm

Stock basis: one known-content M3 kit plus separately stocked M3 x 20 Torx screws supports the two-build comparison on paper. Counts other than the 100 CNC inserts are estimated. Recount before printing; use six Torx screws at accessible crossbrace positions and check their head fit. Inventory is not reserved or consumed.

03 / Shaft and structure

1. Test the shaft/bush and insert interfaces first. Fit the two shaft halves and loose alignment key; install the shaft inserts from the split face before joining the halves with two M3 x 8 screws. Keep every head flush.

2. Fit rotor and two thrust spacers on the central D section. Slide the bushes on from their respective shaft ends. The right end has no enlarged shoulder, so its bush is removable. Leave the rotor clamp light until the plates are assembled.

3. Place the shaft/rotor between P01 and P02. Install both crossbraces with four M3 x 20 screws each. Tighten evenly. Confirm smooth rotation and free axial movement; nominal total axial clearance is 0.4 mm, subject to actual printing.

4. Seat the two stop-pad spigots in P01. Retain each with a recessed M3 x 16 screw. The screw head is flush with the outside plate; the locating spigot transfers transverse stop load. Check both ends of travel gently.

5. Fit the left keyed cam carrier, retaining cap and M3 x 8 screw. Tighten its radial M3 x 10 clamp onto the shaft flat. Keep clearance to the bush flange. Fit a cam with three outward-facing M3 x 10 screws.

6. Attach the grip halves around the keyed stem with two M3 x 20 screws. Tighten the rotor clamp only after checking the full mechanism turns without bearing pinch.

Insert installation

CNC Kitchen M3 inserts: installed OD4.6 mm, length 5.7 mm; nominal pilot 4.0 mm with at least 6.7 mm blind depth. Deeper screw-tip pockets are intentional. Printed compensation must come from a coupon. Install with components separate and allow them to cool before tightening.

Do not tighten a joint until a moving bush becomes a friction clamp. Inspect the actual shaft split, journal roundness and axial clearance. The printed shaft remains split so its long load path can print parallel to the bed.

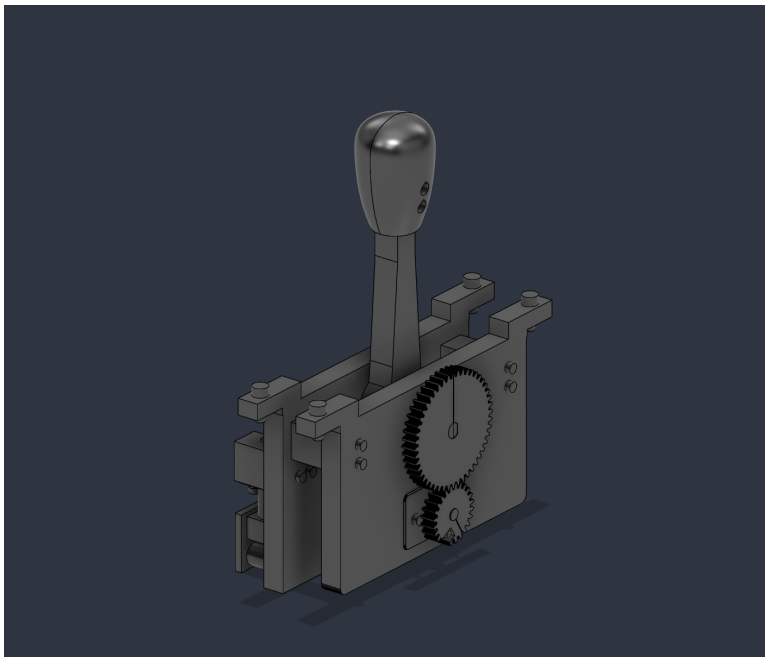
04 / Follower and sensor

7. Fit two 623ZZ bearings between the follower cheeks. The printed pads touch their inner races. Install the M3 x 20 axle and plain nut; the bearings must turn freely after tightening.

8. Fit the two captured follower-pivot halves and the inner axial spacer. Install P33 with its integral compression posts and two recessed M3 x 20 screws. Confirm free follower movement before fitting the spring.

9. Assemble cup, guide sleeves and M3 x 12 guide screws on the reaction block. Retain the spring between cup and follower tail. Attach the reaction block with two M3 x 20 screws. Start with preload fully backed off to its specified position.

10. Mount the pot in the removable panel using its supplied washer/nut. Set both lever and pot to their centers. Install the pinion on the moving 6 mm shaft, then the main gear. Set nominal center distance 45.15 mm; adjust only within 45.0-45.5 mm and verify free mesh.



Actual Fusion model: gear/sensor side. Measured moving shaft length is 8 mm; the additional 7 mm is inferred M7 bushing length. Terminal dimensions and the supplied nut/washer envelope remain unverified.
Mount the unit with four M5 x 14 screws and four 2020 T-nuts. Verify engagement and clearance to the rail web, then route wires away from all moving parts. Never let the pot act as an end stop.

05 / Change cams and commission

1. Back spring preload fully off. Put the lever at its central reference. The eight-gate cam has a central crest; it does not have a central notch.

2. Move the follower to its absolute -4 degree service position. Install the temporary M3 x 12 service screw through the outer plate into the dedicated follower-tail insert. Confirm positive retention before removing the cam screws.

3. Remove the three outward-facing M3 x 10 screws. Withdraw the cam to the left. Leave shaft, carrier, bearing plates, gears and pot in place. Allow at least 20 mm temporary service space; tightly packed modules may need removal from the rails.

4. Fit the replacement cam, tighten its three screws, remove the service screw, then restore preload gradually. Never operate the lever with the service screw fitted.

Physical checks before use

Check	Acceptance to establish physically
Spring	Measure solid height and confirm clearance at every gate, crest and service position. Rate is unrecorded; do not assume the catalog envelope determines force.
Pot	Confirm usable travel exceeds 240 degrees plus margin. Otherwise restrict lever travel before applying force. Verify electrical taper and calibration.
Fits and stops	Check bush rotation, insert retention, shaft/cam/gear clamp slip, stop-pad seating and gentle load response. No impact or rated abuse claim.
Compare with v6	Record assembly/cam-change time, breakaway force, notch-force spread, reversal dead travel, electrical noise and repeat-cycle wear under the same sampling/filtering.

A2:1 gear ratio gives greater sensor rotation for the same lever motion. Effective resolution still depends on pot quality, noise, ADC behavior and backlash.

06 / Print order and orientation

Use the supplied oriented millimeter STLs and your unchanged selected printer/material/process defaults. Print the critical fit interfaces before the large plates. Do not interpret zero generated supports as proof of strength or perfect bridging.

Parts	Bed face / check
P01 / P02	Inner broad faces down. Inspect rail-hole roofs and the 12.3 mm blind stop-socket bridges in P01.
P05 shaft / P25-P26 pivot	Longitudinal split faces down. Inspect insert/key-pocket closures and fit before joining.
P07 rotor	Broad X-min face down; inspect radial clamp roof and lower stop-root continuity.
P08 cams / P32 carrier	Inner X-max face down. Cam counterbores and carrier inserts face upward; inspect carrier pocket floors.
P31 stop pads	Contact X-max face down. Supported tapered base and locating spigot grow upward.
P33 cassette	Outer X-min face down; integral posts grow upward. Inspect 6.2 mm counterbore bridges.
P30 crossbraces	X face down; inspect 4 mm pocket bridges on the lower end.
P13 block / P14 cup	Z-max face down; inspect horizontal mounting roofs and cup pilot.
Other parts	Use the orientation recorded for that STL in mesh-check.json; do not rotate every part by its assembly pose.

Release status

This package supplies prototype geometry, not printer-ready G-code. The existing diagnostic slicer CLI reports machine/tool configuration errors, so its toolpaths must not be sent to the printer. Re-slice the oriented STLs in Bambu Studio with the correct selected printer and filament.

The accompanying VALIDATION.md distinguishes completed digital checks from physical commissioning. Specialist review reports, dispositions and model scripts are retained beside the export for traceability.